TUHIN BANDOPADHYAY

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RESEARCH INTERESTS

Multiphase flows; Particle-laden turbulent flows; Experimental fluid dynamics; Magnetic resonance velocimetry (MRV) and concentration; Urban flows; Biomedical flows

EDUCATION

University of Illinois, Urbana-Champaign Champaign, IL
Doctoral student in Aerospace Engineering. GPA: 4.00/4.00 *Sep. 2020 - Present*

Indian Institute of Technology, Kharagpur kharagpur, India
B.Tech+M.Tech (Dual degree) in Aerospace Engineering. GPA: 9.50/10 *Sep. 2015 - July 2020*

AWARDS AND FELLOWSHIPS

Graduate College Block Grant Fellowship, Grainger College of Engineering, UIUC	2025
Best Technical Accomplishment Award, poster competition, Aerospace Engineering, UIUC	2025
Mavis Future Faculty Fellowship, Grainger College of Engineering, UIUC	2024-2025
Graduate College Block Grant Fellowship, Grainger College of Engineering, UIUC	2024
Faculty Endowed Graduate Award, Aerospace Engineering, UIUC	2024
Best Project Award for Bachelor and Master Project, IIT Kharagpur	2020
ASME IGTI Student Scholarship, International Gas Turbine Institute, ASME	2018-2019
Summer Research Fellowship, Indian Academy of Science	2018
MCM Scholarship, IIT Kharagpur	2015-2018
JBNSTS Fellowship, Jagadish Bose National Science Talent Search	2015-2016
Academic Excellence Award, Office of Chief Minister, West Bengal, India	2015

RESEARCH EXPERIENCE

Dynamics of inertial particle clusters in turbulent flows Aug. 2020 - Present
Graduate Research Assistant, Aerospace Engineering Department, UIUC *Advisor: Dr. Laura Villafañe*

- Designed, developed, and characterized a particle-laden turbulent channel flow facility (the largest in the USA) to study the coupled interactions of inertial particles with turbulence experimentally.
- Developed a novel and computationally efficient method for particle cluster identification and temporal tracking to analyze coherent particle dynamics in particle-laden turbulent flows.
- Investigating experimentally the coherent dynamics of the inertial particles in turbulent flows for varying parameter space.

Turbulent scalar dispersion measurement using MRC Oct. 2023 - Present
Project Lead, Aerospace Engineering Department, UIUC *Advisors: Dr. Laura Villafañe, Dr. Brad Sutton*

- Conducted experimental investigations into scalar dispersion within urban environments, focusing on dispersion patterns around model building arrays.
- Utilized Magnetic Resonance Concentration (MRC) measurements to obtain three-dimensional concentration fields around buildings.

Effect of wind direction on the ventilation dynamics of a model sports stadium April 2023 - Present
Project Lead, UIUC and West Point *Advisors: Dr. Laura Villafañe, Dr. Andrew J. Banko*

- Investigated Experimentally scalar dispersion around a model stadium as a prototypical open-air structure.
- Combined novel medical imaging techniques with laser-based diagnostics to obtain a comprehensive set of flow velocity and concentration fields.
- Measured external and internal three-dimensional velocity fields using magnetic resonance velocimetry, while time-resolved concentration fields at a central plane are obtained using planar laser-induced fluorescence.

Three-dimensional velocity measurements in periodic flows via MRV April 2021 - March 2022
Project Lead, Aerospace Engineering & Beckman Institute, UIUC Advisors: Dr. Laura Villafañe, Dr. Brad Sutton

- Benchmarked the performance of the MRI machine, developed the post-processing tools, and established the optimal experimental framework to scan complex 3-dimensional periodic transient flows.
- Participated in the Magnetic Resonance Velocimetry (MRV) Challenge, 2021.

Characterization of the leading-edge separation bubble June 2018 - June 2020
Master's Thesis Research, Aerospace Engineering Department, IIT Kharagpur Advisor: Dr. Chetan S. Mistry

- Designed and developed a wind tunnel test section to study separated boundary layer characteristics over the leading edge of a flat plate, with the provision to change the angle of incidence.
- Studied detailed flow physics inside leading-edge separation bubble for varying Reynolds numbers and angle of incidences using 2D particle image velocimetry.

Institute challenge research project: 'Airavat' May 2017 - May 2020
Founding Student Leader, Aerospace Engineering Department, IIT Kharagpur Advisor: Dr. Chetan S. Mistry

- Secured funding of 1 million Rupees from the Institute Innovation Grant through a competitive selection process to develop a flying car propelled by green energy (electric).
- Designed and fabricated the vehicle's propulsion system using axial flow ducted fans, serving as the head of the propulsion team.

Instabilities in low Reynolds number round jets May 2018 - July 2018
Summer Intern, Aerospace Engineering Department, IIT Kanpur Advisor: Dr. Debopam Das

- Investigated experimentally the behavior of a non-buoyant circular water jet discharged from a circular nozzle for Reynolds number range of $150 - 1000$.
- Performed dye flow visualization and 2D PIV to understand the instability mechanisms before jet breakdown.

JOURNAL PUBLICATIONS

6. (*Under preparation*) **Tuhin Bandopadhyay**, Laura Villafañe, Andrew J. Banko. "Effect of Wind Direction on the Ventilation Dynamics of a Model Sports Stadium." *Experiments in Fluids*, 2025.
5. (*Under preparation*) **Tuhin Bandopadhyay**, Laura Villafañe. "Characterization of vertical particle-laden channel flow facility for varying particle and flow parameters." *Experiments in Fluids*, 2025.
4. **Tuhin Bandopadhyay**, Alvaro Tomas Gil, Laura Villafañe. "Inertial particle cluster dynamics in the core region of a turbulent duct flow." *International Journal of Multiphase Flow*, 2025.
3. Benson M., Banko A., Elkins C., An D.G., Song S, Bruschewski M., Grundmann S., **Bandopadhyay T.**, Sutton B., Villafane L., Han K., Hwang W., Eaton J.K. "MRV Challenge 2: Phase locked turbulent measurements in a roughness array." *Experiments in Fluids, 64(28)*, 2023.
2. **Bandopadhyay, Tuhin**, Chetan S. Mistry. "Characterization of Leading-edge Laminar Separation Bubble for various Angles of Incidence and Reynolds Numbers." *Journal of The Institution of Engineers (India): Series C*, 2022.
1. **Bandopadhyay, Tuhin**, Chetan S. Mistry. "Effects of Total Pressure Distribution on Performance of Small-Size Counter-Rotating Axial-Flow Fan Stage for Electrical Propulsion." *ASME Open Journal of Engineering 1*, 2022.

CONFERENCE PUBLICATIONS/TALKS

9. **Tuhin Bandopadhyay**, Laura Villafañe. "Statistical and time-resolved analysis of inertial particle clusters in channel flow turbulence." *12th International Conference on Multiphase flow, Toulouse, France*, 2025.
8. Andrew J. Banko, **Tuhin Bandopadhyay**, Brad Sutton, Laura Villafañe. "Three-dimensional flow structure and ventilation dynamics of a cross-ventilated building." *Bulletin of the American Physical Society*, 2024.
7. **Tuhin Bandopadhyay**, Laura Villafañe. "Time-resolved measurements of clustered particles in channel flow turbulence." *Bulletin of the American Physical Society*, 2024.
6. **Tuhin Bandopadhyay**, Laura Villafañe. "Unraveling the persistence and dynamics of high-concentrated regions of inertial particles in turbulent flows." *Bulletin of the American Physical Society*, 2023.
5. Villafañe, L., **Bandopadhyay T.**, Sutton, B., Ellwein, B., Gehl, J., Iliff, S., ... & Banko, A. " Effect of Wind Direction on the Ventilation Dynamics of a Model Sports Stadium." *Bulletin of the American Physical Society*, 2023.
4. **Tuhin Bandopadhyay**, Laura Villafañe. "Coherence of inertial particle clusters in the core region of a turbulent channel flow." *Bulletin of the American Physical Society 67*, 2022.
3. **Tuhin Bandopadhyay**, Laura Villafañe. "Tracking particle clusters in turbulent channel flows via density-based clustering algorithms." *APS Division of Fluid Dynamics Meeting Abstracts*, 2021.
2. Villafane, L., **Bandopadhyay, T.**, Tafti, S., & Sutton, B. "Three-dimensional velocity measurements in pulsatile flows via Magnetic Resonance Velocimetry, 2021 MRV Challenge." *APS Division of Fluid Dynamics Meeting Abstracts*, 2021.
1. **Tuhin Bandopadhyay**, Chetan S. Mistry. "Effects of Total Pressure Distribution on Performance of Small Size Counter-Rotating Axial-Flow Fan Stage for Electrical Propulsion." *ASME Gas Turbine India Conference*, 2019.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant — Fall 2024, Fall 2023, Fall 2022, Spring 2022
Incompressible flow (AE311) — *Aerospace Engineering, UIUC*

- Taught course modules, held office hours, helped develop academic instructional materials (HWs, projects), created lab demonstration videos, and graded HWs and projects.

Research mentoring — 2021-Present
Undergraduate and graduate students mentor — *Aerospace Engineering, UIUC*

- Mentoring one undergraduate student for semester-long AE298 research project in Spring 2025.
- Mentored two graduate students toward their Master's thesis.
- Mentored seven undergraduate students to perform semester-long research projects in our research laboratory.

TECHNICAL SKILLS

Experimental techniques
PIV, PTV, Hot-wire anemometry, MRV, MRC, pressure, and temperature sensors, load-cell

Programming
C, C++, Python, Matlab, Labview, Arduino

Software
Fluent, CFX, ICEM CFD, Turbogrid, Solidworks, Catia, Tecplot

Operating system
Windows, Linux based operating systems (Ubuntu, Raspbian)